CCNA Practice Final 2

Implementación de redes de área amplia y servicios distribuidos (TC3003B.570)

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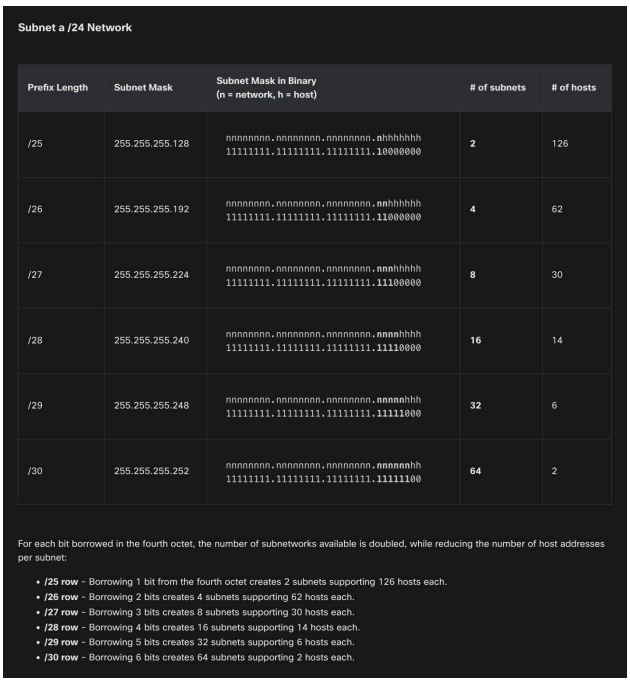
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1. Three devices are on three different subnets. Match the network address and the broadcast address with each subnet where these devices are located. Device 1: IP address 192.168.10.77/28 on subnet 1. Device 2: IP address 192.168.10.17/30 on subnet 2. Device 3: IP address 192.168.10.35/29 on subnet 3.
   - **Respuesta: Subnet 1 network number -> 192.168.10.64; Subnet 2 broadcast address -> 192.168.10.19; Subnet 3 network number -> 192.168.10.32; Subnet 1 broadcast address -> 192.168.10.79; Subnet 2 network number -> 192.168.10.16; Subnet 3 broadcast address -> 192.168.10.39**
   - La tabla de división en subredes muestra que tomar bits prestados en el cuarto octeto duplica las subredes y reduce la cantidad de direcciones de host: `/28` crea bloques de 16 direcciones, `/29` bloques de 8 direcciones y `/30` bloques de 4 direcciones.
   - Para la subred 1 (`/28`), la dirección `.77` se ubica en el intervalo `.64` a `.79`, por lo que la dirección de red es `192.168.10.64` y la de difusión es `192.168.10.79`.
   - Para la subred 2 (`/30`), la dirección `.17` se ubica en el intervalo `.16` a `.19`, con dirección de red `192.168.10.16` y difusión en `192.168.10.19`.
   - Para la subred 3 (`/29`), la dirección `.35` se ubica en el intervalo `.32` a `.39`, con dirección de red `192.168.10.32` y difusión en `192.168.10.39`.
2. Given network 172.18.109.0, which subnet mask would be used if 6 host bits were available?
   - **Respuesta: 255.255.255.192**
   - La tabla oficial de subredes de una red `/24` indica que la fila `/26` reserva exactamente 6 bits para la porción de host en el cuarto octeto.
   - En notación binaria, el cuarto octeto con 2 bits de red y 6 bits de host (`11000000`) equivale en formato decimal punteado a `255.255.255.192`.
   - Esta máscara `/26` genera 4 subredes utilizables y permite albergar 62 direcciones de host por subred.
3. Refer to the exhibit. A user issues the command netstat -r on a workstation. Which IPv6 address is one of the link-local addresses of the workstation?
   - **Respuesta: fe80::30d0:115:3f57:fe4c/128**
   - El material de Cisco define que las direcciones IPv6 link local residen exclusivamente en el rango reservado `fe80::/10`.
   - Los primeros 10 bits fijos (`1111 1110 10xx xxxx`) restringen el primer hexteto a un valor comprendido entre `fe80` y `febf`.
   - En la salida del comando `netstat -r`, la dirección `fe80::30d0:115:3f57:fe4c/128` contiene el prefijo `fe80` y la longitud `/128` que identifica la interfaz de host local en ese enlace.
4. What type of IPv6 address is represented by ::1/128?
   - **Respuesta: loopback**
   - El diagrama de clasificación de direcciones IPv6 unicast distingue categorías como Global Unicast, Link local, Loopback, Unspecified, Unique Local y Embedded IPv4.
   - Dentro de este esquema formal, la dirección `::1/128` corresponde a la dirección de loopback local.
   - El nodo utiliza `::1/128` para enviar paquetes hacia sí mismo y comprobar que la pila de protocolos TCP/IP funcione correctamente sin enviar tráfico al medio físico.

5. Refer to the exhibit. What does the value of the window size specify?
   - **Respuesta: the amount of data that can be sent before an acknowledgment is required**
   - El texto de control de flujo en TCP determina que el tamaño de ventana define la cantidad de bytes que pueden transmitirse antes de esperar un acuse de recibo.
   - Este valor corresponde al volumen de datos que el equipo receptor puede aceptar y almacenar en su búfer en un instante específico.
   - El tamaño de ventana se negocia durante el establecimiento de la sesión y el receptor lo actualiza de forma continua en cada segmento transmitido.
6. To which TCP port group does the port 414 belong?
   - **Respuesta: well-known**
   - La IANA divide el espacio de puertos de 16 bits (0 a 65535) en 3 rangos: puertos conocidos de 0 a 1023, puertos registrados de 1024 a 49151 y puertos privados o dinámicos de 49152 a 65535.
   - Los puertos conocidos se reservan para servicios y aplicaciones universales como servidores web, transferencia de archivos y acceso remoto.
   - Dado que el número 414 se ubica dentro del rango de 0 a 1023, queda clasificado como un puerto conocido.
7. Which statement describes the characteristics of packet-filtering and stateful firewalls as they relate to the OSI model?
   - **Respuesta: A packet-filtering firewall typically can filter up to the transport layer, whereas a stateful firewall can filter up to the session layer.**
   - El material de Cisco explica que los firewalls de filtrado de paquetes inspeccionan direcciones IP y puertos analizando tráfico hasta la capa de transporte (Capa 4).
   - Los firewalls con inspección de estado monitorean las conexiones activas y verifican que el tráfico entrante corresponda a respuestas legítimas hasta la capa de sesión (Capa 5).
   - Al conservar el estado de cada flujo, el firewall stateful bloquea paquetes entrantes no solicitados y mitiga ataques directos a la red.
8. Only employees connected to IPv6 interfaces are having difficulty connecting to remote networks. The analyst wants to verify that IPv6 routing has been enabled. What is the best command to use to accomplish the task?
   - **Respuesta: show running-config**
   - Para que un router Cisco encamine paquetes entre subredes IPv6 es obligatorio habilitar el comando de configuración global `ipv6 unicast-routing`.
   - La forma directa de confirmar que esta instrucción global forma parte de la configuración activa del router consiste en consultar el archivo con `show running-config`.
   - Otros comandos informan el estado operativo de interfaces individuales pero no validan si el motor de enrutamiento global IPv6 está activo en el dispositivo.
9. What happens when a switch receives a frame and the calculated CRC value is different than the value that is in the FCS field?
   - **Respuesta: The switch drops the frame.**
   - El campo de secuencia de verificación de trama (FCS) de 4 bytes permite detectar si ocurrieron errores de bits durante el trayecto físico.
   - El switch receptor recalcula el código de redundancia cíclica (CRC) y lo compara contra el valor registrado en el campo FCS al final de la trama.
   - Si ambos valores no coinciden, el switch determina que la trama contiene señales no deseadas o datos dañados y descarta la trama de inmediato.
10. Which two issues can cause both runts and giants in Ethernet networks? (Choose two.)
    - **Respuesta: a malfunctioning NIC; half-duplex operations**
    - El estándar Ethernet establece que cualquier trama menor a 64 bytes se cataloga como fragmento de colisión o trama enana, producto habitual de colisiones en enlaces semidúplex.
    - Las tramas que sobrepasan el tamaño máximo esperado de 1518 bytes se consideran tramas gigantes y suelen generarse por una tarjeta de red (NIC) defectuosa que transmite señales corruptas.
    - Por lo tanto, operar en modo semidúplex con colisiones frecuentes y una tarjeta de red con mal funcionamiento son las 2 causas de tramas enanas y gigantes.
11. Which two functions are performed at the LLC sublayer of the OSI Data Link Layer to facilitate Ethernet communication? (Choose two.)

- **Respuesta: places information in the Ethernet frame that identifies which network layer protocol is being encapsulated by the frame; adds Ethernet control information to network protocol data**
- La subcapa de control de enlace lógico (LLC, IEEE 802.2) enlaza el software de red de los niveles superiores con los componentes de hardware de los niveles inferiores.
- La subcapa LLC inserta información en la trama que especifica cuál protocolo de Capa 3 (como IPv4 o IPv6) viaja encapsulado dentro del paquete de datos.
- Además, la subcapa LLC agrega información de control de enlace a los datos del protocolo de red antes de transferirlos a la subcapa MAC.

Anexo: Capturas del material de Cisco Introduction to Networks

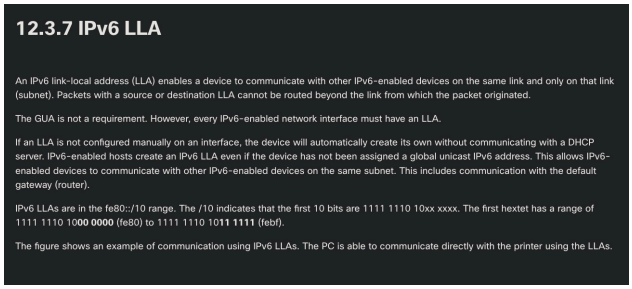


Subnet a /24 Network

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.**h**hhhhhhh 11111111.11111111.11111111.**1**0000000	**2**	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nn**hhhhhh 11111111.11111111.11111111.**11**000000	**4**	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnn**hhhhh 11111111.11111111.11111111.**111**00000	**8**	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnn**hhhh 11111111.11111111.11111111.**1111**0000	**16**	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnn**hhh 11111111.11111111.11111111.**11111**000	**32**	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnnn**hh 11111111.11111111.11111111.**111111**00	**64**	2

For each bit borrowed in the fourth octet, the number of subnetworks available is doubled, while reducing the number of host addresses per subnet:

- **/25 row** - Borrowing 1 bit from the fourth octet creates 2 subnets supporting 126 hosts each.
- **/26 row** - Borrowing 2 bits creates 4 subnets supporting 62 hosts each.
- **/27 row** - Borrowing 3 bits creates 8 subnets supporting 30 hosts each.
- **/28 row** - Borrowing 4 bits creates 16 subnets supporting 14 hosts each.
- **/29 row** - Borrowing 5 bits creates 32 subnets supporting 6 hosts each.
- **/30 row** - Borrowing 6 bits creates 64 subnets supporting 2 hosts each.



12.3.7 IPv6 LLA

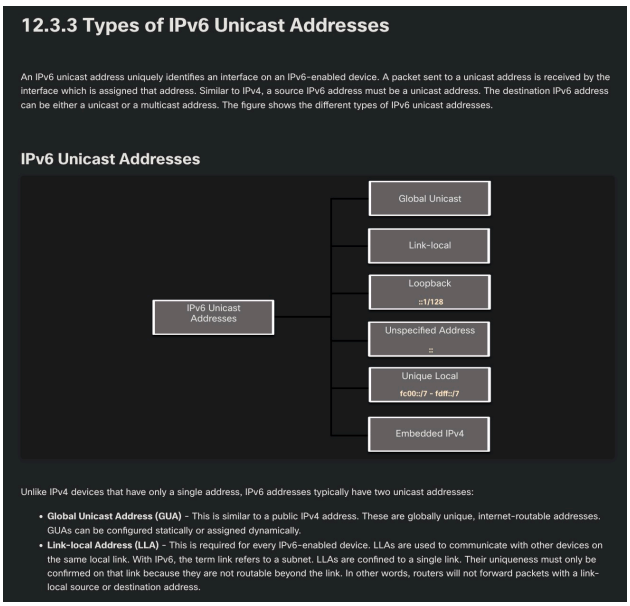
An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet). Packets with a source or destination LLA cannot be routed beyond the link from which the packet originated.

The GUA is not a requirement. However, every IPv6-enabled network interface must have an LLA.

If an LLA is not configured manually on an interface, the device will automatically create its own without communicating with a DHCP server. IPv6-enabled hosts create an IPv6 LLA even if the device has not been assigned a global unicast IPv6 address. This allows IPv6-enabled devices to communicate with other IPv6-enabled devices on the same subnet. This includes communication with the default gateway (router).

IPv6 LLAs are in the fe80::/10 range. The /10 indicates that the first 10 bits are 1111 1110 10xx xxxx. The first hextet has a range of 1111 1110 10**00 0000** (feb0) to 1111 1110 10**11 1111** (febf).

The figure shows an example of communication using IPv6 LLAs. The PC is able to communicate directly with the printer using the LLAs.



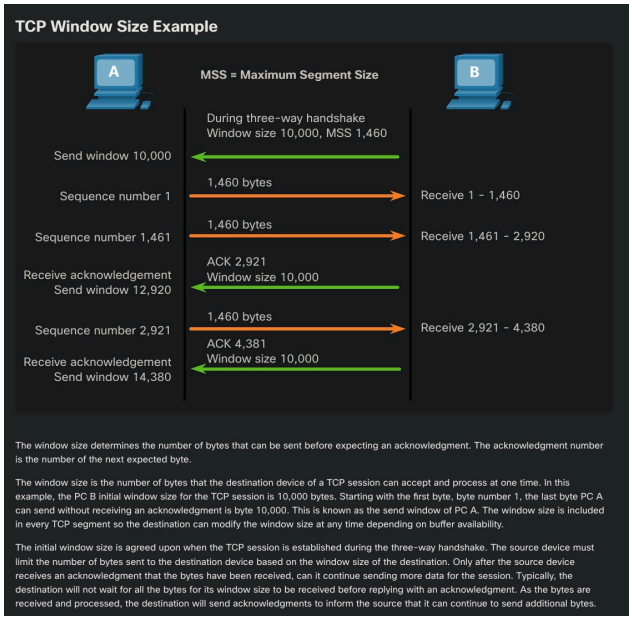
12.3.3 Types of IPv6 Unicast Addresses

An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device. A packet sent to a unicast address is received by the interface which is assigned that address. Similar to IPv4, a source IPv6 address must be a unicast address. The destination IPv6 address can be either a unicast or a multicast address. The figure shows the different types of IPv6 unicast addresses.

IPv6 Unicast Addresses

Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:

- **Global Unicast Address (GUA)** - This is similar to a public IPv4 address. These are globally unique, internet-routable addresses. GUAs can be configured statically or assigned dynamically.
- **Link-local Address (LLA)** - This is required for every IPv6-enabled device. LLAs are used to communicate with other devices on the same local link. With IPv6, the term link refers to a subnet. LLAs are confined to a single link. Their uniqueness must only be confirmed on that link because they are not routable beyond the link. In other words, routers will not forward packets with a link-local source or destination address.



TCP Window Size Example

The window size determines the number of bytes that can be sent before expecting an acknowledgment. The acknowledgment number is the number of the next expected byte.

The window size is the number of bytes that the destination device of a TCP session can accept and process at one time. In this example, the PC B initial window size for the TCP session is 10,000 bytes. Starting with the first byte, byte number 1, the last byte PC A can send without receiving an acknowledgment is byte 10,000. This is known as the send window of PC A. The window size is included in every TCP segment so the destination can modify the window size at any time depending on buffer availability.

The initial window size is agreed upon when the TCP session is established during the three-way handshake. The source device must limit the number of bytes sent to the destination device based on the window size of the destination. Only after the source device receives an acknowledgment that the bytes have been received, can it continue sending more data for the session. Typically, the destination will not wait for all the bytes for its window size to be received before replying with an acknowledgment. As the bytes are received and processed, the destination will send acknowledgments to inform the source that it can continue to send additional bytes.

14.4.3 Port Number Groups

The Internet Assigned Numbers Authority (IANA) is the standards organization responsible for assigning various addressing standards, including the 16-bit port numbers. The 16 bits used to identify the source and destination port numbers provides a range of ports from 0 through 65535.

The IANA has divided the range of numbers into the following three port groups.

Port Group	Number Range	Description
Well-known Ports	**0 to 1,023**	• These port numbers are reserved for common or popular services and applications such as web browsers, email clients, and remote access clients. • Defined well-known ports for common server applications enables clients to easily identify the associated service required.
Registered Ports	**1,024 to 49,151**	• These port numbers are assigned by IANA to a requesting entity to use with specific processes or applications. • These processes are primarily individual applications that a user has chosen to install, rather than common applications that would receive a well-known port number. • For example, Cisco has registered port 1812 for its RADIUS server authentication process.
Private and/or **Dynamic Ports**	**49,152 to 65,535**	• These ports are also known as *ephemeral ports*. • The client's OS usually assign port numbers dynamically when a connection to a service is initiated. • The dynamic port is then used to identify the client application during communication.

16.3.6 Types of Firewalls

Firewall products come packaged in various forms. These products use different techniques for determining what will be permitted or denied access to a network. They include the following:

- **Packet filtering** - Prevents or allows access based on IP or MAC addresses
- **Application filtering** - Prevents or allows access by specific application types based on port numbers
- **URL filtering** - Prevents or allows access to websites based on specific URLs or keywords
- **Stateful packet inspection (SPI)** - Incoming packets must be legitimate responses to requests from internal hosts. Unsolicited packets are blocked unless permitted specifically. SPI can also include the capability to recognize and filter out specific types of attacks, such as denial of service (DoS)

7.1.4 Ethernet Frame Fields

The minimum Ethernet frame size is 64 bytes and the expected maximum is 1518 bytes. This includes all bytes from the destination MAC address field through the frame check sequence (FCS) field. The preamble field is not included when describing the size of the frame.

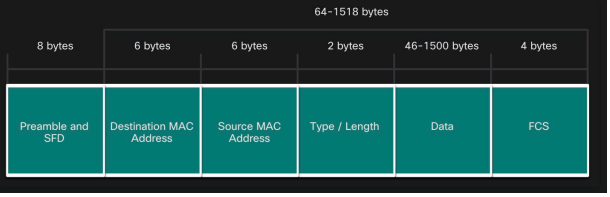
Note: The frame size may be larger if additional requirements are included, such as VLAN tagging. VLAN tagging is beyond the scope of this course.

Any frame less than 64 bytes in length is considered a "collision fragment" or "runt frame" and is automatically discarded by receiving stations. Frames with more than 1500 bytes of data are considered "jumbo" or "baby giant frames".

If the size of a transmitted frame is less than the minimum, or greater than the maximum, the receiving device drops the frame. Dropped frames are likely to be the result of collisions or other unwanted signals. They are considered invalid. Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.

The figure shows each field in the Ethernet frame. Refer to the table for more information about the function of each field.

Ethernet Frame Fields



12.4.1 Static GUA Configuration on a Router

As you learned in the previous topic, IPv6 GUAs are the same as public IPv4 addresses. They are globally unique and routable on the IPv6 internet. An IPv6 LLA lets two IPv6-enabled devices communicate with each other on the same link (subnet). It is easy to statically configure IPv6 GUAs and LLAs on routers to help you create an IPv6 network. This topic teaches you how to do just that!

Most IPv6 configuration and verification commands in the Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of **ipv6** in place of **ip** within the commands.

For example, the Cisco IOS command to configure an IPv4 address on an interface is **ip address** *ip-address subnet-mask*. In contrast, the command to configure an IPv6 GUA on an interface is **ipv6 address** *ipv6-address/prefix-length*.

Notice that there is no space between *ipv6-address* and *prefix-length*.

The example configuration uses the topology shown in the figure and these IPv6 subnets:

- 2001:db8:acad:1::/64
- 2001:db8:acad:2::/64
- 2001:db8:acad:3::/64

7.1.2 Data Link Sublayers

IEEE 802 LAN/MAN protocols, including Ethernet, use the following two separate sublayers of the data link layer to operate. They are the Logical Link Control (LLC) and the Media Access Control (MAC), as shown in the figure.

Recall that LLC and MAC have the following roles in the data link layer:

- **LLC Sublayer** – This IEEE 802.2 sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers. It places information in the frame that identifies which network layer protocol is being used for the frame. This information allows multiple Layer 3 protocols, such as IPv4 and IPv6, to use the same network interface and media.
- **MAC Sublayer** – This sublayer (IEEE 802.3, 802.11, or 802.15 for example) is implemented in hardware and is responsible for data encapsulation and media access control. It provides data link layer addressing and is integrated with various physical layer technologies.